

Safety of Electronic Control Gear for LED Modules (LED Drivers)

CRS Compulsory Registration Scheme - Electronics Standard

IS 15885 : 2021

1 Scope

1.1 This standard specifies safety and performance requirements for electronic control gear (drivers) for LED modules and LED lamps intended for use with supply voltages up to 1000 V AC or 1500 V DC at frequencies up to 1000 Hz.

1.2 This standard applies to LED drivers for general lighting purposes in indoor and outdoor luminaires. It covers both constant-voltage (CV) and constant-current (CC) driver types.

1.3 This standard forms part of the Compulsory Registration Scheme (CRS) for electronics and IT goods under the Bureau of Indian Standards Act 2016. Compliance is mandatory before placing products on the Indian market.

NOTE LED modules and luminaires containing integral drivers are covered in IS 10322 (Part 5). This standard applies only to the driver unit tested as a separate component.

2 Normative References

IS 302 (Part 1) : 2008 - Safety of household and similar electrical appliances: General Requirements.

IS 15111 : 2012 - Safety of transformers, reactors, power supply units and combinations - General requirements and tests.

IS 16024 : 2012 - LED modules for general lighting - Safety specifications.

IEC 61347-1 : 2015 - Lamp controlgear - Part 1: General and safety requirements.

3 Definitions

3.1 LED Driver - An electronic device that regulates the power delivered to an LED module or LED array; it may incorporate dimming, thermal management, and power factor correction.

3.2 Constant Current (CC) Driver - A driver that maintains a constant output current over a specified range of output voltage.

3.3 Constant Voltage (CV) Driver - A driver that maintains a constant output voltage over a specified range of output current.

3.4 Rated Output Current - The nominal value of the output current declared by the manufacturer.

3.5 Power Factor (PF) - The ratio of active power consumed to the apparent power; the minimum PF is specified in Table 1 according to rated output power.

4 Classification

4.1 LED drivers shall be classified by: output type (CC or CV); rated output power bracket; protection class (Class I, II, or III); and degree of protection (IP20, IP44, IP65, IP67).

5 General Requirements

5.1 Marking and Documentation

5.1.1 Each LED driver shall be permanently and legibly marked with: supply voltage and frequency; rated output current or voltage; protection class symbol; IP rating; manufacturer name or trademark; and ISI mark confirming compliance with this standard.

5.1.2 The BIS Registration Number obtained under the CRS Order shall be clearly marked on the driver or its immediate packaging.

5.2 Electrical Safety Requirements

5.2.1 Dielectric strength - LED drivers shall withstand a dielectric strength test at 1500 V AC (or 2125 V DC) for 1 minute between input and output circuits without breakdown.

5.2.2 Insulation resistance - Insulation resistance between live parts and accessible metal parts shall not be less than 2 MOhm when measured with a 500 V DC instrument.

5.2.3 Earth continuity - For Class I constructions, the resistance between the earth terminal and any accessible metal part shall not exceed 0.1 Ohm.

6 Performance Requirements

6.1 Efficiency and Power Factor

6.1.1 The minimum efficiency and power factor shall comply with the values given in Table 1 at rated input voltage and rated output load.

Table 1 Minimum Efficiency and Power Factor Requirements (IS 15885:2021)

Rated Output Power (W)	Min. Efficiency (%)	Min. Power Factor	Test Condition
5 or less	70	0.50	Full load, 230 V, 50 Hz
More than 5 to 10	75	0.70	Full load, 230 V, 50 Hz
More than 10 to 25	80	0.90	Full load, 230 V, 50 Hz
More than 25 to 50	85	0.90	Full load, 230 V, 50 Hz
More than 50 to 100	87	0.92	Full load, 230 V, 50 Hz
More than 100	88	0.95	Full load, 230 V, 50 Hz

6.2 Output Regulation

6.2.1 Current regulation - For CC drivers, the output current shall not deviate by more than 5 percent of the rated output current when the input voltage varies from 90 percent to 110 percent of rated supply

voltage.

6.2.2 Voltage regulation - For CV drivers, the output voltage shall not deviate by more than 5 percent of the rated output voltage across 0 to 100 percent of rated output current.

6.3 Thermal Requirements

6.3.1 The temperature at any accessible surface of the driver shall not exceed the limits given in Table 2 during steady-state operation at rated conditions.

Table 2 Maximum Accessible Surface Temperature Limits (IS 15885:2021)

Surface / Location	Max Temperature (deg C)	Test Method
Metal surfaces (unintentionally touchable)	60	Clause 8.3
Non-metal surfaces (unintentionally touchable)	75	Clause 8.3
Surfaces intended to be touched	48	Clause 8.3
Internal winding - Class E	120	Clause 8.4
Internal winding - Class B	130	Clause 8.4
Internal winding - Class F	155	Clause 8.4
Internal winding - Class H	180	Clause 8.4

6.4 Electromagnetic Compatibility (EMC)

6.4.1 Conducted emission limits for Class B equipment as per IS 13252 (Part 1) shall be met when tested over the frequency range 150 kHz to 30 MHz.

6.4.2 Harmonic current limits shall comply with IS 16604 (equivalent to IEC 61000-3-2 Class C) for rated active input power above 25 W.

7 Compulsory Registration Scheme (CRS) Requirements

7.1 Registration obligation - LED drivers covered by this standard fall under the Electronics and Information Technology Goods (Compulsory Registration) Order. No such product may be imported, sold, or stocked for sale in India without a valid BIS Registration Certificate.

7.2 Testing laboratory - All tests required for registration shall be conducted at a BIS-recognised test laboratory or NABL-accredited laboratory with the relevant scope.

7.3 Factory inspection - BIS shall carry out factory inspections and market surveillance as deemed necessary. The registered applicant shall maintain quality records as specified in Annex B for a minimum period of five years.

7.4 Self-Declaration of Conformity (SDoC) based on this standard does NOT satisfy the CRS requirement; mandatory third-party certification by BIS is required.

8 Test Methods

8.1 Sample size - Unless otherwise specified, tests shall be performed on a minimum of three samples representative of the production batch.

8.2 Input conditions - Tests shall be carried out at 230 V +/- 2 percent, 50 Hz +/- 1 percent unless otherwise specified in the relevant test clause.

8.3 Surface temperature measurement - Measure using calibrated thermocouples or calibrated non-contact infrared thermometer at thermal steady state (temperature variation less than 1 deg C over 30 min).

8.4 Winding temperature measurement - Determine winding temperature by the resistance method as specified in IS 15111.